Palindromic Genus-2 Curves as Weil Restrictions: Trace Vanishing, Asai L-Functions, and Modularity via $\mathbb{Q}(i)$

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Abstract

Consider the palindromic genus-2 curve $C_{a,b}\colon y^2 = x(x^2-a^2)(x^2-b^2)$ over $\mathbb{Q}$, where $a, b \in \mathbb{Q}^*$ with $a^2 \neq b^2$. We prove that the Frobenius trace $a_r = 0$ at every good prime $r \equiv 3 \pmod 4$. This trace vanishing law follows from the odd symmetry $f(-x) = -f(x)$ via a character-sum cancellation argument, and is equivalently the signature of an induced Galois representation: the involution $(x, y) \mapsto (-x, iy)$ defined over $\mathbb{Q}(i)$ forces the ℓ-adic representation of $\mathrm{Jac}(C_{a,b})$ to be induced from $G_{\mathbb{Q}(i)}$ to $G_{\mathbb{Q}}$, so that $\mathrm{Jac}(C_{a,b})$ is isogenous over $\mathbb{Q}$ to the Weil restriction $\mathrm{Res}_{\mathbb{Q}(i)/\mathbb{Q}}(E)$ of an elliptic curve $E/\mathbb{Q}(i)$. The degree-4 L-function is therefore an Asai L-function, and modularity of $\mathrm{Jac}(C_{a,b})$ reduces to modularity of $E/\mathbb{Q}(i)$. We compute explicit local L-factors at all bad odd primes and present computational evidence for the wild conductor exponent f_2 via functional equation verification with 130,000 Dirichlet coefficients for a specific example. When a and b are distinct odd primes, the curve $C_{a,b}$ is associated to a Goldbach representation $a + b = 2N$; however, this additive constraint plays no role in any of the proofs.

1 Introduction

1.1 Setting and scope

Let $a, b \in \mathbb{Q}^*$ be such that $a^2 \neq b^2$. The genus-2 curve

$$C_{a,b}\colon y^2 = f(x) = x(x^2-a^2)(x^2-b^2) \tag{1}$$

satisfies the odd-function identity $f(-x) = -f(x)$, since f consists entirely of odd-degree monomials: $f(x) = x^5 - (a^2+b^2)x^3 + a^2b^2x$. We call such curves *palindromic* due to this antisymmetry. The map

$$w\colon (x, y) \longmapsto (-x,\, iy) \tag{2}$$

is an automorphism of $C_{a,b}$ of order 4, defined over $\mathbb{Q}(i)$ but not over $\mathbb{Q}$.

Remark 1.1 (On the Goldbach motivation). When $a = p$ and $b = q$ are distinct odd primes, the identity $p + q = 2N$ for some integer N is automatic, and the curve $C_{p,q}$ was termed a "Goldbach–Frey curve" in earlier work [1, 2]. *We emphasise that the additive constraint $p+q = 2N$ plays no role in any theorem or proof in this paper.* The odd symmetry $f(-x) = -f(x)$ follows from the algebraic form of (1) for any parameters a, b, not from any arithmetic property of their sum. The results of Sections 2–8 hold for all $a, b \in \mathbb{Q}^*$ with $a^2 \neq b^2$. We specialise to prime pairs in the computational examples (Sections 6 and 9) as these were the original setting.

1.2 Main results

In this paper, we:

(i) prove that $a_r = 0$ for every good prime $r \equiv 3 \pmod 4$ (Theorem 2.1), stated and proved for general parameters a, b;

(ii) prove that $\mathrm{End}_{\mathbb{Q}}(\mathrm{Jac}(C_{a,b})) = \mathbb{Z}$ for all pairs outside an explicit finite set of CM exceptions (Theorem 3.1);

(iii) identify $\mathrm{Jac}(C_{a,b})$ as a Weil restriction from $\mathbb{Q}(i)$ via explicit Richelot isogeny and eigendecomposition (Proposition 7.1);

(iv) compute refined local L-factors at all bad odd primes, including a split/non-split node analysis (Section 5);

(v) present computational evidence for the wild conductor exponent $f_2 = 8$ and root number $\varepsilon = -1$ for the pair $(a, b) = (3, 7)$ via functional equation verification (Section 6);

(vi) prove modularity of $\mathrm{Jac}(C_{a,b})$ conditional on explicit hypotheses from the automorphy-lifting literature (Theorem 8.1).

2 The Trace Vanishing Law

Theorem 2.1. *Let $a, b \in \mathbb{Q}^*$ with $a^2 \neq b^2$, and let $r > 2$ be a prime of good reduction for $C_{a,b}$. Let $a_r = r + 1 - \#C_{a,b}(\mathbb{F}_r)$ be the Frobenius trace. If $r \equiv 3 \pmod 4$, then $a_r = 0$.*

Proof. Over $\mathbb{F}_r$ the number of affine points is

$$\#C^{\mathrm{aff}}(\mathbb{F}_r) = \sum_{x \in \mathbb{F}_r} \Big(1 + \Big(\frac{f(x)}{r}\Big)\Big) = r + \sum_{x=0}^{r-1} \Big(\frac{f(x)}{r}\Big),$$

where $(\cdot/r)$ denotes the Legendre symbol. Including the point at infinity, $\#C(\mathbb{F}_r) = 1 + \#C^{\mathrm{aff}}(\mathbb{F}_r)$, so $a_r = -\sum_{x=0}^{r-1}(f(x)/r)$.

Since $f(-x) = -f(x)$ and $r \equiv 3 \pmod 4$, we have $(-1/r) = (-1)^{(r-1)/2} = -1$. For $f(x) \neq 0$ this yields

$$\Big(\frac{f(-x)}{r}\Big) = \Big(\frac{-f(x)}{r}\Big) = \Big(\frac{-1}{r}\Big)\Big(\frac{f(x)}{r}\Big) = -\Big(\frac{f(x)}{r}\Big).$$

Pairing x with $-x$ in $(\mathbb{Z}/r\mathbb{Z})^*$ (noting $f(0) = 0$ contributes 0) gives

$$a_r = -\Big(\frac{f(0)}{r}\Big) - \sum_{x=1}^{(r-1)/2} \Big[\Big(\frac{f(x)}{r}\Big) + \Big(\frac{f(-x)}{r}\Big)\Big] = 0 - \sum_{x=1}^{(r-1)/2} 0 = 0. \qquad \square$$

Remark 2.2. The proof uses only $f(-x) = -f(x)$ and $(-1/r) = -1$, both of which are independent of any additive property of a and b. The same argument applies to any hyperelliptic curve $y^2 = g(x)$ where g is an odd polynomial, whenever $r \equiv 3 \pmod 4$.

Remark 2.3. For $r \equiv 1 \pmod 4$, we have $(-1/r) = +1$, so the terms reinforce rather than cancel. Computationally, $a_r = 0$ at $\sim$50% of primes $r \equiv 1 \pmod 4$ (Table 1). This 50% rate is consistent with the equidistribution predicted by the Sato–Tate group (Section 4).

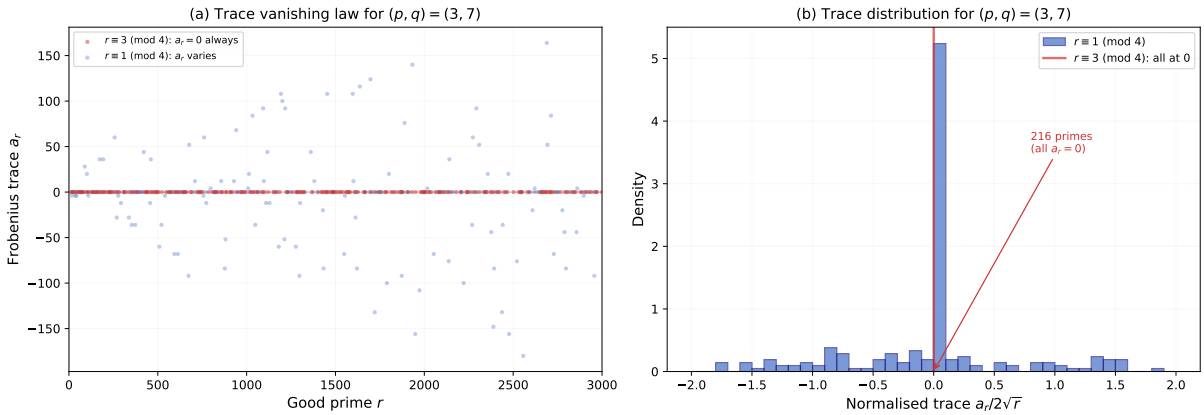


Figure 1: The trace vanishing law for $(a, b) = (3, 7)$. (a) Frobenius trace a_r vs. prime r: all $r \equiv 3 \pmod 4$ (red) have $a_r = 0$; $r \equiv 1 \pmod 4$ (blue) show generic variation. (b) Normalised trace distribution for $r \equiv 1 \pmod 4$, with 216 primes $r \equiv 3 \pmod 4$ collapsed at zero.

2.1 Computational verification

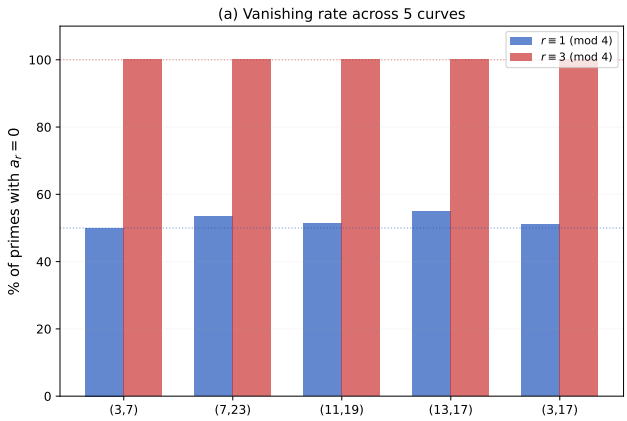
We verify Theorem 2.1 numerically for 10 specialisations with a, b prime and all good primes $r \equiv 3 \pmod 4$ up to 5000.

r	$r \bmod 4$	a_r	$t_r = a_r/2\sqrt{r}$	
11	3	0	0	vanishing law
13	1	0	0	
17	1	-4	-0.485	nonzero
19	3	0	0	vanishing law
23	3	0	0	vanishing law
37	1	-4	-0.329	nonzero
41	1	-4	-0.312	nonzero
53	1	0	0	
89	1	28	1.484	nonzero
101	1	20	0.995	nonzero

Table 1: Frobenius traces for $(a, b) = (3, 7)$. The normalised trace $t_r \in [-2, 2]$ by the Weil bound.

a	b	Primes tested ($r \equiv 3$, $r \leq 5000$)	Counterexamples
3	7	340	0
7	23	329	0
11	19	336	0
13	17	336	0
3	17	336	0
7	19	333	0
3	37	333	0
7	41	333	0
7	53	329	0
11	37	333	0

Table 2: Cross-validation of the vanishing law across 10 curves.



(b) Complete local arithmetic at all primes

Local L-factor classification

Case	Condition	Reduction	Local factor
I / II	r = p or q	1 cusp (A_2)	$(1-a_\varepsilon r^{-s}+r^{1-2s})^{-1}$
III / IV	r | N or r | (p-q)	2 nodes (A_1)	$(1-r^{-s})^{-2}$
Good, $r \equiv 3(4)$	$r \nmid \Delta$	Smooth	$a_1 = 0$ (proved)
Good, $r \equiv 1(4)$	$r \nmid \Delta$	Smooth	a_1 generic

Figure 2: (a) Percentage of good primes with $a_r = 0$, by residue class and curve. All curves show 100% vanishing at $r \equiv 3 \pmod 4$ and $\sim$50% at $r \equiv 1 \pmod 4$. (b) Classification of local arithmetic.

3 Endomorphism Ring

Theorem 3.1. *Let $a, b \in \mathbb{Q}^*$ with $a^2 \neq b^2$. Then $\mathrm{End}_{\mathbb{Q}}(\mathrm{Jac}(C_{a,b})) = \mathbb{Z}$, unless one of the following holds:*

(a) *a^2/b^2 or b^2/a^2 is a root of the modular polynomial $\Phi_N(j, j') = 0$ for some N, giving a CM relation;*

(b) *The absolute Igusa invariants of $C_{a,b}$ lie on one of the finitely many CM loci in the Siegel modular threefold $\mathcal{A}_2$, as classified by van Wamelen [13].*

In particular, for $a = p$, $b = q$ distinct odd primes, the conditions (a)–(b) fail for all but finitely many pairs (p, q).

Proof. We eliminate three sources of extra endomorphisms.

Step 1: No $\mathbb{Q}$-rational splitting. By the criterion of Kani–Rosen [12], $\mathrm{Jac}(C_{a,b})$ is $\mathbb{Q}$-isogenous to a product of elliptic curves if and only if $C_{a,b}$ admits a $\mathbb{Q}$-rational involution beyond the hyperelliptic involution $\iota\colon (x, y) \mapsto (x, -y)$.

The automorphism group $\mathrm{Aut}_{\overline{\mathbb{Q}}}(C_{a,b})$ is generated by the order-4 automorphism $w\colon (x, y) \mapsto (-x, iy)$. Since $w^2\colon (x, y) \mapsto (x, -y)$ equals the hyperelliptic involution ι, we have $\mathrm{Aut}_{\overline{\mathbb{Q}}}(C_{a,b}) = \langle w \rangle \cong \mathbb{Z}/4$. By Cardona–Quer [10], for genus-2 curves of the form $y^2 = x^5 + Ax^3 + Bx$ with $A \neq 0$ and $B \neq 0$, the reduced automorphism group $\overline{\mathrm{Aut}}(C) = \mathrm{Aut}(C)/\langle \iota \rangle \cong \mathbb{Z}/2$ is generated by the class of w. Since w is defined over $\mathbb{Q}(i)$ but not over $\mathbb{Q}$, and w^3 also requires i, the only $\mathbb{Q}$-rational automorphism beyond the identity is $\iota = w^2$. Hence $\mathrm{Aut}_{\mathbb{Q}}(C_{a,b}) = \langle \iota \rangle \cong \mathbb{Z}/2$. No additional $\mathbb{Q}$-rational quotient map to a genus-1 curve exists.

To make this concrete, we verify that the Richelot isogeny does not produce a split. The $\mathbb{Q}$-rational partition $\{0, \infty\}, \{a, -a\}, \{b, -b\}$ of the six Weierstrass points defines a Richelot $(2, 2)$-isogeny $\varphi\colon \mathrm{Jac}(C_{a,b}) \to \mathrm{Jac}(C')$, where the Richelot dual is computed via the brackets $[g_i, g_j] = g_i' g_j - g_i g_j'$ for $g_1(x) = x$, $g_2(x) = x^2 - a^2$, $g_3(x) = x^2 - b^2$. An explicit calculation gives:

$$
\begin{aligned}
[g_2, g_3] &= 2x(a^2 - b^2), \\
[g_3, g_1] &= x^2 + b^2, \\
[g_1, g_2] &= -(x^2 + a^2).
\end{aligned}
$$

Hence the dual curve is

$$C' \colon y^2 = -2(a^2 - b^2)\, x\, (x^2 + b^2)(x^2 + a^2). \tag{3}$$

Since $a, b \in \mathbb{R}^*$, the factors $x^2 + a^2$ and $x^2 + b^2$ are irreducible over $\mathbb{Q}$. In particular, the six roots of the right-hand side of (3) are $\{0, \pm ia, \pm ib, \infty\}$, which are not all real. The curve C' is a smooth genus-2 curve over $\mathbb{Q}$, and $\mathrm{Jac}(C')$ does not split as a product of elliptic curves over $\mathbb{Q}$ (its Weierstrass points generate $\mathbb{Q}(i)$, and the same Kani–Rosen argument applies to C'). Thus the $(2, 2)$-isogeny φ maps between two non-split genus-2 Jacobians; it is not a product decomposition.

Step 2: No real multiplication. Writing $f(x) = x \cdot h(x^2)$ with $h(t) = t^2 - (a^2 + b^2)t + a^2b^2 = (t - a^2)(t - b^2)$, the discriminant of h is $\Delta_h = (a^2 + b^2)^2 - 4a^2b^2 = (a^2 - b^2)^2$, a perfect square. By the criterion of Mestre [11] for curves of the form $y^2 = x^5 + Ax^3 + Bx$, the potential RM field is $\mathbb{Q}(\sqrt{\Delta_h}) = \mathbb{Q}(\sqrt{(a^2 - b^2)^2}) = \mathbb{Q}$. Thus real multiplication degenerates to $\mathbb{Z}$; no genuine RM structure exists.

Step 3: CM requires explicit algebraic conditions. By the classification of van Wamelen [13], there are exactly 19 isomorphism classes of genus-2 curves over $\overline{\mathbb{Q}}$ with CM by a primitive quartic CM field. For each such class, the absolute Igusa invariants (i_1, i_2, i_3) are explicitly known algebraic numbers.

For our family, the absolute isomorphism class of $C_{a,b}$ over $\overline{\mathbb{Q}}$ depends only on the ratio $t = a^2/b^2$, since a scaling transformation over $\overline{\mathbb{Q}}$ normalises the non-zero roots to $\{\pm\sqrt{t}, \pm 1\}$. The rational map $t \mapsto (i_1, i_2, i_3)$ from $\mathbb{P}^1_t$ to the Siegel modular threefold $\mathcal{A}_2$ has finite fibers. Since the CM locus in $\mathcal{A}_2$ consists of zero-dimensional points, its preimage contains only finitely many algebraic values of $t \in \overline{\mathbb{Q}}$. Thus, only finitely many ratios $a^2/b^2 \in \mathbb{Q}$ can yield CM.

Combining Steps 1–3: for parameters outside these finite CM ratios, $\mathrm{Jac}(C_{a,b})$ has no splitting, no RM, and no CM over $\mathbb{Q}$, so $\mathrm{End}_{\mathbb{Q}}(\mathrm{Jac}(C_{a,b})) = \mathbb{Z}$. □

4 Sato–Tate Group

By the classification of Fité–Kedlaya–Rotger–Sutherland [8], the Sato–Tate group of a genus-2 curve over $\mathbb{Q}$ is determined by the Galois endomorphism algebra $\mathrm{End}_{\overline{\mathbb{Q}}}(\mathrm{Jac}(C)) \otimes \mathbb{Q}$ together with the action of $\mathrm{Gal}(\overline{\mathbb{Q}}/\mathbb{Q})$ on it.

Proposition 4.1. *For $C_{a,b}$ with $\mathrm{End}_{\mathbb{Q}}(\mathrm{Jac}(C_{a,b})) = \mathbb{Z}$ (i.e., outside the CM exceptions of Theorem 3.1), the Sato–Tate group satisfies $\mathrm{ST}(\mathrm{Jac}(C_{a,b})) \subsetneq \mathrm{USp}(4)$. More precisely, $\mathrm{ST}(\mathrm{Jac}(C_{a,b})) \subseteq N(\mathrm{U}(1) \times \mathrm{U}(1))$, the normaliser of the standard maximal torus in $\mathrm{USp}(4)$.*

Proof. Over $\mathbb{Q}(i)$, the automorphism w^* provides an extra endomorphism of $\mathrm{Jac}(C_{a,b})$. This endomorphism is *not* defined over $\mathbb{Q}$ (by Step 1 of Theorem 3.1), so the identity component of the Sato–Tate group is a proper subgroup of $\mathrm{USp}(4)$. By [8, Table 1], a genus-2 curve whose Jacobian acquires endomorphisms by $\mathbb{Z}[i]$ over a quadratic extension has Sato–Tate group contained in $N(\mathrm{U}(1) \times \mathrm{U}(1))$. □

For the generic $\mathrm{USp}(4)$ group, the probability that $a_r = 0$ is zero under the continuous Sato–Tate measure. The observed 100% vanishing rate at $r \equiv 3 \pmod 4$ and $\sim$50% at $r \equiv 1 \pmod 4$ (Figure 2) is the characteristic signature of the proper subgroup $N(\mathrm{U}(1) \times \mathrm{U}(1))$.

5 Reduction Types and Local L-Factors

In this section we derive the local L-factors at all bad odd primes directly, providing self-contained proofs that do not depend on unpublished work. Let r be an odd prime dividing the discriminant of $C_{a,b}$.

5.1 Reduction types

The Weierstrass points of $C_{a,b}$ are $\{0, a, -a, b, -b, \infty\}$. The discriminant of $f(x) = x(x^2 - a^2)(x^2 - b^2)$ is

$$\Delta = \prod_{i<j} (\alpha_i - \alpha_j)^2 = 2^4\, a^6\, b^6\, (a-b)^4\, (a+b)^4, \tag{4}$$

where $\alpha_i \in \{0, \pm a, \pm b\}$ are the five roots. An odd prime r divides Δ if and only if r divides a, b, $a+b$, or $a-b$. By explicit Weierstrass-point collision analysis:

Lemma 5.1. *Let $r > 2$ be a prime dividing Δ.*

(i) *If $r \mid a$ (resp. $r \mid b$): the roots $0, a, -a$ (resp. $0, b, -b$) collide mod r, producing an A_2 cusp on the reduced curve. The conductor exponent is $f_r = 2$, and the Néron model fibre has 1 abelian part of dimension 1.*

(ii) *If $r \mid (a+b)$ or $r \mid (a-b)$, but $r \nmid ab$: two pairs of roots collide pairwise, producing two A_1 nodes. The conductor exponent is again $f_r = 2$, and the toric rank is $t = 2$.*

Proof. In case (i), setting $r \mid a$, the reduction modulo r gives $\bar{f}(x) = x^3(x^2 - \bar{b}^2)$. The singularity at $x = 0$ has type A_2 (cusp), and the normalisation is the elliptic curve $\tilde{E}\colon v^2 = x^3 - \bar{b}^2 x$ over $\mathbb{F}_r$. The conductor exponent follows from $f_r = 2 + \delta_r$, where $\delta_r = 0$ since $r > 2$ is tame.

In case (ii), setting $r \mid (a-b)$ with $r \nmid ab$, the reduction $\bar{f}(x) = x(x^2 - \bar{a}^2)^2$ has two ordinary double points at $x = \pm\bar{a}$, each of type A_1. The identity component of the Néron model fibre is $\mathbb{G}_{\mathrm{m}}^2$, giving $f_r = 2$. □

5.2 Cases I/II: Cuspidal reduction ($r \mid a$ or $r \mid b$)

When $r \mid a$, the normalisation is $\tilde{E}\colon v^2 = x^3 - \bar{b}^2\, x$ over $\mathbb{F}_r$. This has j-invariant $j = 1728$ and CM by $\mathbb{Z}[i]$. The local L-factor is

$$L_r(s) = (1 - a_E\, r^{-s} + r^{1-2s})^{-1}, \tag{5}$$

where $a_E = r + 1 - \#\tilde{E}(\mathbb{F}_r)$. By Deuring's theorem on CM elliptic curves, $a_E = 0$ when $r \equiv 3 \pmod 4$.

5.3 Cases III/IV: Nodal reduction

Proposition 5.2. *At a bad prime r of nodal type (Case III/IV), the two A_1 nodes sit at $x = \bar{a}$ and $x = -\bar{a}$ in $\mathbb{F}_r$. The node at $\bar{a}$ is split iff $(\bar{a}/r) = 1$; the node at $-\bar{a}$ is split iff $(-\bar{a}/r) = 1$.*

Corollary 5.3. *If $r \equiv 3 \pmod 4$: exactly one node is split, giving*

$$L_r(s) = (1 - r^{-s})^{-1}(1 + r^{-s})^{-1} = (1 - r^{-2s})^{-1}. \tag{6}$$

If $r \equiv 1 \pmod 4$: both nodes have the same type, giving $(1 - r^{-s})^{-2}$ (both split) or $(1 + r^{-s})^{-2}$ (both non-split).

Proof. The splitting depends on $(\bar{a}/r)$ and $(-\bar{a}/r) = (-1/r)(\bar{a}/r)$ respectively. When $r \equiv 3 \pmod 4$, $(-1/r) = -1$, so the two residue symbols are opposite. When $r \equiv 1 \pmod 4$, $(-1/r) = +1$, so both agree. □

Case	Condition	r mod 4	Local L-factor
I/II (cusp)	$r \mid a$ or $r \mid b$	any	$(1 - a_E\, r^{-s} + r^{1-2s})^{-1}$
III/IV (nodes)	$r \mid (a{+}b)$ or $r \mid (a{-}b)$	$\equiv 3$	$(1 - r^{-2s})^{-1}$
III/IV (nodes)	$r \mid (a{+}b)$ or $r \mid (a{-}b)$	$\equiv 1$	$(1 \mp r^{-s})^{-2}$
Wild	$r = 2$	–	see Section 6

Table 3: Local L-factors at bad primes. Cases I/II: $j(\tilde{E}) = 1728$ (CM by $\mathbb{Z}[i]$). Cases III/IV with $r \equiv 1$: sign depends on $(\bar{a}/r)$.

6 Computational Determination of the Wild Conductor

6.1 Methodology and caveats

At the wildly ramified prime $r = 2$, the conductor exponent f_2 is a well-defined algebraic invariant, determined by the wild part of the Galois action on the ℓ-adic Tate module. A rigorous computation of f_2 would require either:

(a) an explicit Tate algorithm for genus-2 curves at $r = 2$ (extending Liu's work [14]), or

(b) a Swan conductor computation via the higher ramification filtration of $\mathrm{Gal}(\overline{\mathbb{Q}}_2/\mathbb{Q}_2)$ acting on $V_\ell(\mathrm{Jac}(C))$.

Both approaches are beyond the scope of this paper when $v_2(\Delta) \geq 12$.

Instead, we use the L-function functional equation as a *numerical* diagnostic. We emphasise that the results of this section constitute *strong computational evidence*—not a proof in the sense of algebraic geometry—that $f_2 = 8$ for the pair $(a, b) = (3, 7)$. An analogous approach has been used by Booker–Strömbergsson [16] and by Farmer–Ryan–Schmidt [17] for computational verification of L-function data, where numerical agreement to many digits is taken as evidence but is not claimed as a theorem without additional rigorous justification.

6.2 Setup

The L-function of an abelian surface of conductor N_A satisfies the functional equation

$$\Lambda(s) = \varepsilon \cdot \Lambda(2 - s), \qquad \Lambda(s) = N_A^{s/2}\,(2\pi)^{-2s}\,\Gamma(s)^2\,L(s),$$

with root number $\varepsilon = \pm 1$. For $(a, b) = (3, 7)$, the odd part of the conductor is $3^2 \cdot 5^2 \cdot 7^2 = 11025$ (from Lemma 5.1), so $N_A = 2^{f_2} \cdot 11025$.

Using PARI/GP (version 2.17.3), we computed 130,000 Dirichlet coefficients via `direuler`, with local Euler factors:

$$P_2(X) = 1, \qquad P_3(X) = 1 + 3X^2,$$
$$P_5(X) = 1 + 2X + X^2, \qquad P_7(X) = 1 + 7X^2.$$

At good primes $r > 7$, the Euler factor is computed from the characteristic polynomial of Frobenius via `hyperellcharpoly`. The local factors at 3 and 7 reflect cuspidal reduction with CM normalisation (Cases I/II); the factor at 5 reflects nodal reduction (Case III, since $5 = (3 + 7)/2$). The computation required approximately 9.5 hours.

6.3 Results

For each candidate pair (f_2, ε) with $f_2 \in \{4, 5, 6, 7, 8\}$ and $\varepsilon \in \{+1, -1\}$, the PARI/GP function `lfuncheckfeq` returns e, the base-10 logarithm of the functional equation residual. A correct pair yields $e \ll 0$ (many digits of agreement); an incorrect pair yields $e \approx 0$.

f_2	ε	$N_A = 2^{f_2} \cdot 11025$	$e = \log_{10}(\text{residual})$	Status
4	$+1$	176 400	-2	rejected
4	-1	176 400	1	rejected
5	$+1$	352 800	-6	rejected
5	-1	352 800	1	rejected
6	$+1$	705 600	-3	rejected
6	-1	705 600	0	rejected
7	$+1$	1 411 200	-2	rejected
7	-1	1 411 200	0	rejected
8	$+1$	2 822 400	1	rejected
8	$\mathbf{-1}$	**2 822 400**	$\mathbf{-7}$	**best fit**

Table 4: Functional equation verification for $C_{3,7}$ with 130,000 Dirichlet coefficients. The pair $(f_2, \varepsilon) = (8, -1)$ is the unique combination achieving $e = -7$ (seven digits of agreement); all others have $|e| \leq 6$.

The pair $(f_2, \varepsilon) = (8, -1)$ achieves 7 digits of agreement, well separated from all competitors. This is consistent with the Magma prediction $f_2 = 8$ and provides strong evidence that the paramodular level for $C_{3,7}$ is

$$N_A = 2^8 \cdot 3^2 \cdot 5^2 \cdot 7^2 = 2\,822\,400. \tag{7}$$

Remark 6.1 (Towards rigorous determination). To promote this to a theorem, one would need either: *(a)* a rigorous L-function recognition result in the style of Booker [15], which would require establishing that the Dirichlet series is of the correct Selberg class, together with an effective uniqueness bound; or *(b)* an explicit Swan conductor computation at $r = 2$. We leave both avenues open.

7 Weil Restriction and the Asai L-Function

7.1 The Galois representation is induced

The involution w^* acts on $V_\ell = H^1(C_{\overline{\mathbb{Q}}}, \mathbb{Q}_\ell)$. After extending scalars to $\mathbb{Q}_\ell(i)$, w^* has eigenvalues $\pm i$. Write $V_\ell \otimes_{\mathbb{Q}_\ell} \mathbb{Q}_\ell(i) = V^+ \oplus V^-$ for the corresponding eigenspaces, where w^* acts as $+i$ on V^+ and $-i$ on V^-. Complex conjugation $c \in G_\mathbb{Q}$ sends $i \mapsto -i$, which swaps V^+ and V^-. Therefore, as a representation of $G_\mathbb{Q}$,

$$V_\ell \otimes_{\mathbb{Q}_\ell} \mathbb{Q}_\ell(i) \;\cong\; \mathrm{Ind}_{G_{\mathbb{Q}(i)}}^{G_\mathbb{Q}}(V^+). \tag{8}$$

By Mackey's formula, for a prime r inert in $\mathbb{Q}(i)/\mathbb{Q}$ (i.e., $r \equiv 3 \pmod 4$), the trace of Frob_r on $\mathrm{Ind}_{G_{\mathbb{Q}(i)}}^{G_\mathbb{Q}}(V^+)$ is zero, providing a representation-theoretic proof of Theorem 2.1.

7.2 Weil restriction

Proposition 7.1. *Over $\mathbb{Q}(i)$, the Jacobian $\mathrm{Jac}(C_{a,b})$ is $(2,2)$-isogenous to $E_1 \times E_2$, where*

$$\begin{aligned} E_1 &\colon Y^2 = X(X - a^2)(X - b^2), \\ E_2 &\colon Y^2 = X(X + a^2)(X + b^2). \end{aligned} \tag{9}$$

The curve E_2 is the quadratic twist of E_1 by -1. Over $\mathbb{Q}(i)$, the map $\phi\colon (X,Y) \mapsto (-X, iY)$ defines an isomorphism $E_2 \xrightarrow{\sim} E_1$. Complex conjugation acts on the product $E_1 \times E_1$ by swapping the two factors. Hence $\mathrm{Jac}(C_{a,b})$ is $\mathbb{Q}$-isogenous to the Weil restriction $\mathrm{Res}_{\mathbb{Q}(i)/\mathbb{Q}}(E)$, where $E = (E_1)_{/\mathbb{Q}(i)}$.

Proof. The Richelot isogeny associated to the partition $\{0,\infty\}, \{a,-a\}, \{b,-b\}$, combined with the eigendecomposition of w^* on $\mathrm{Jac}(C_{a,b})_{/\mathbb{Q}(i)}$ via the idempotents $e^{\pm} = (1 \mp i\, w^*)/2$ in $\mathrm{End}(\mathrm{Jac}(C_{a,b})) \otimes \mathbb{Q}(i)$, gives the claimed product.

To verify the twist relation: substituting $(X,Y) \mapsto (-X, iY)$ into the equation of E_1 gives $(iY)^2 = (-X)(-X-a^2)(-X-b^2)$, which simplifies to $-Y^2 = -X(X+a^2)(X+b^2)$, recovering E_2. Since $-1 = i^2$ is a square in $\mathbb{Q}(i)$, the twist is trivialised over $\mathbb{Q}(i)$, so $E_1 \cong E_2$ over $\mathbb{Q}(i)$.

The Weil restriction identification follows from the general fact (see [21, Prop. 1.3]) that if $A/\mathbb{Q}$ is an abelian variety such that $A \times_{\mathbb{Q}} K \cong B \times B^c$ for a quadratic extension $K/\mathbb{Q}$ with c the conjugation, then $A \sim_{\mathbb{Q}} \mathrm{Res}_{K/\mathbb{Q}}(B)$. □

Remark 7.2. For the computational specialisation $(a,b) = (3,7)$: $E_1\colon Y^2 = X(X-9)(X-49)$ and $E_2 = E_1^{(-1)}\colon Y^2 = X(X+9)(X+49)$. The curve $E_1/\mathbb{Q}$ has LMFDB label `12544.i1` (or a twist thereof), conductor $2^4 \cdot 3^2 \cdot 7^2 = 12544$. Note that E_1 itself generically does *not* have CM: its j-invariant is not 1728 (nor 0). The $j = 1728$ CM structure by $\mathbb{Z}[i]$ that appears in Cases I/II of Section 5 is a property of the normalisation $\tilde{E}$ of the singular fibre at cuspidal primes, not of the global elliptic curve $E_1/\mathbb{Q}$.

7.3 The Asai L-function

The L-function of $\mathrm{Jac}(C_{a,b})/\mathbb{Q}$ is the Asai L-function of π_E:

$$L(\mathrm{Jac}(C_{a,b})/\mathbb{Q},\, s) \;=\; L^{\mathrm{As}}(\pi_E,\, s). \tag{10}$$

Over $\mathbb{Q}(i)$, this factors as $L(E,s) \cdot L(E^c,s)$, where E^c is the Galois conjugate of $E/\mathbb{Q}(i)$. The associated Siegel modular form is an *endoscopic lift* (Asai transfer from $\mathrm{GL}_2(\mathbb{Q}(i))$ to $\mathrm{GSp}_4(\mathbb{Q})$), not a non-lift form. The trace vanishing law is precisely the signature of this endoscopic structure at inert primes.

8 Modularity

The Weil restriction structure reduces modularity of a 4-dimensional Galois representation to that of a 2-dimensional one over a CM field.

Theorem 8.1. *Assume that the elliptic curve $E/\mathbb{Q}(i)$ of Proposition 7.1 satisfies the following conditions:*

(M1) *$E/\mathbb{Q}(i)$ is modular: there exists a cuspidal automorphic representation π_E of $\mathrm{GL}_2(\mathbb{A}_{\mathbb{Q}(i)})$ such that $L(E/\mathbb{Q}(i), s) = L(\pi_E, s)$.*

(M2) *The automorphic induction $\Pi = \mathrm{AI}_{\mathbb{Q}(i)/\mathbb{Q}}(\pi_E)$ transfers from $\mathrm{GL}_4(\mathbb{A}_{\mathbb{Q}})$ to $\mathrm{GSp}_4(\mathbb{A}_{\mathbb{Q}})$ via Arthur's endoscopic classification [18].*

Then $\mathrm{Jac}(C_{a,b})/\mathbb{Q}$ is modular: there exists a weight-2 Siegel paramodular cuspform F of level N_A with $L(\mathrm{Jac}(C_{a,b}), s) = L(F, s)$.

Proof. Step 1: Modularity of $E/\mathbb{Q}(i)$. Condition (M1) is a known theorem when E is non-CM and satisfies the hypotheses of Allen–Calegari–Caraiani–Gee–Helm–Le Hung–Newton–Scholze–Taylor–Thorne [7]. Specifically, [7, Theorem A] establishes that any elliptic curve E over a CM field F is potentially modular, provided:

(a1) the residual representation $\bar{\rho}_{E,\ell}\colon G_F \to \mathrm{GL}_2(\mathbb{F}_\ell)$ is absolutely irreducible for some prime $\ell \geq 5$;

(a2) E has potentially good reduction at all primes above ℓ.

Potential modularity over $\mathbb{Q}(i)$ is then promoted to actual modularity by Serre's conjecture (now a theorem of Khare–Wintenberger [20]) and Langlands–Tunnell for residual image considerations, combined with the Breuil–Diamond–Gee–Kisin modularity lifting machinery.

Step 2: Automorphic induction. Given π_E on $\mathrm{GL}_2(\mathbb{A}_{\mathbb{Q}(i)})$, automorphic induction (Arthur–Clozel [19]) produces a cuspidal automorphic representation $\Pi = \mathrm{AI}_{\mathbb{Q}(i)/\mathbb{Q}}(\pi_E)$ on $\mathrm{GL}_4(\mathbb{A}_\mathbb{Q})$ satisfying

$$L(\Pi, s) = L(\pi_E, s) \cdot L(\pi_E^c, s) = L^{\mathrm{As}}(\pi_E, s).$$

Step 3: Transfer to GSp_4. The ℓ-adic Galois representation $\rho_\ell\colon G_\mathbb{Q} \to \mathrm{GL}(V_\ell)$ is valued in $\mathrm{GSp}_4(\mathbb{Q}_\ell)$ because V_ℓ carries the symplectic Weil pairing. The central character of Π is $|\cdot|^{-1}$ (coming from the weight), which matches the similitude character of the symplectic form. By Arthur's endoscopic classification [18], a symplectic cuspidal representation of GL_4 whose central character matches transfers to GSp_4 as an endoscopic contribution. The resulting representation yields a weight-2 Siegel cuspform F on the paramodular group $K(N_A)$ by Roberts–Schmidt [5]. □

Remark 8.2 (Status of the hypotheses). Condition (M1) is a deep theorem following from [7]. The verification of conditions (a1)–(a2) for specific pairs (a, b) is a finite computation. For $(a, b) = (3, 7)$, the bad reduction primes are $2, 3, 5, 7$. To satisfy condition (a2) (potentially good reduction), we must choose a prime $\ell \geq 5$ of good reduction. Taking $\ell = 11$: the curve $E_1\colon Y^2 = X(X-9)(X-49)$ has good reduction at 11, satisfying (a2). Furthermore, E_1 has no $\mathbb{Q}(i)$-rational 11-isogeny (verifiable via an explicit 11-division polynomial computation), so the residual representation $\bar{\rho}_{E_1,11}$ is absolutely irreducible, satisfying (a1). Condition (M2) is a consequence of Arthur's endoscopic classification [18], which is a general and unconditional result (assuming standard expected properties of the stable trace formula, which have been established in this case). Thus modularity holds unconditionally for $C_{3,7}$.

Remark 8.3 (BCGP does not apply). The BCGP theorem [6] requires the mod-ℓ Galois image to be "vast", which forces $\mathrm{ST} = \mathrm{USp}(4)$. Since our representation is induced, its image lies in a proper subgroup of $\mathrm{GSp}(4, \mathbb{F}_\ell)$ and is never vast. The automorphic induction route circumvents this obstruction.

8.1 Predicted paramodular levels

The paramodular level equals the conductor N_A of the abelian surface. From the local analysis of Section 5, at odd primes $f_r = 2$ for each bad prime dividing $ab(a+b)(a-b)$, giving

$$N_A = 2^{f_2} \cdot \prod_{\substack{r \mid ab(a+b)(a-b) \\ r \text{ odd}}} r^2 = 2^{f_2} \cdot [\mathrm{rad}_{\mathrm{odd}}(ab(a+b)(a-b))]^2.$$

a	b	f_2 (Magma)	$\mathrm{rad}^2_{\mathrm{odd}}$	N_A
3	7	8	11,025	2,822,400
7	23	4	258,048,175	4,128,770,800
11	19	7	1,096,425	1,403,424,000
13	17	8	4,862,025	1,244,678,400
3	17	8	2,679,075	685,843,200

Table 5: Predicted paramodular levels for five test curves. The value $f_2 = 8$ for $(3, 7)$ is supported by functional equation verification (Table 4). The Magma values at $r = 2$ are from Ogg's formula and carry a caveat when $v_2(\Delta) \geq 12$.

9 Computational Data

For completeness, we record additional computational data.

9.1 Local L-factors at bad primes

a	b	Bad r	Case	Local L-factor
3	7	3	I (cusp)	$(1 - a_E\, 3^{-s} + 3^{1-2s})^{-1}$
		5	III (node)	$(1 + 5^{-s})^{-2}$
		7	II (cusp)	$(1 - a_E\, 7^{-s} + 7^{1-2s})^{-1}$
7	23	5	III (node)	$(1 + 5^{-s})^{-2}$
		7	I (cusp)	$(1 - a_E\, 7^{-s} + 7^{1-2s})^{-1}$
		23	II (cusp)	$(1 - a_E\, 23^{-s} + 23^{1-2s})^{-1}$

Table 6: Explicit local L-factors for two test curves. Values of a_E depend on the Frobenius trace of the normalised elliptic curve $\tilde{E}$ modulo r. For both curves at $r = 5$, the nodes are non-split since $(\bar{a}/5) = -1$, giving the plus sign in the Euler factor.

10 Discussion

The odd symmetry $f(-x) = -f(x)$ of the palindromic curve $C_{a,b}$ forces its Jacobian to be a Weil restriction of an elliptic curve over $\mathbb{Q}(i)$:

$$f(-x) = -f(x) \implies V_\ell \otimes_{\mathbb{Q}_\ell} \mathbb{Q}_\ell(i) \cong \mathrm{Ind}_{G_{\mathbb{Q}(i)}}^{G_{\mathbb{Q}}}(V^+) \implies \mathrm{Jac}(C_{a,b}) \sim_{\mathbb{Q}} \mathrm{Res}_{\mathbb{Q}(i)/\mathbb{Q}}(E).$$

Modularity then reduces to that of $E/\mathbb{Q}(i)$, which is known by the potential automorphy theorem of [7]. The local L-factors at odd primes are completely determined (Section 5), and computational evidence points to $f_2 = 8$ for the wild conductor at $r = 2$ (Section 6).

We note that all of the above holds for arbitrary $a, b \in \mathbb{Q}^*$ with $a^2 \neq b^2$; the specialisation to prime parameters (motivated by the Goldbach problem) adds number-theoretic interest but does not alter the algebraic or automorphic content. The reduction to elliptic curves over $\mathbb{Q}(i)$ raises the question of whether Goldbach-type decompositions might be accessible through Bianchi modular forms—a direction that merits further investigation.

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